Enhanced LSTM-Based Time Series Forecasting of Transformer Oil Temperature: A Comparative Study with Attention and Transformer Architectures

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***Abstract*—Accurate forecasting of transformer oil temperature supports condition monitoring and predictive maintenance in electric power systems. This paper presents a controlled comparison of three deep learning forecasters on the ETTh1 benchmark: a two-layer baseline Long Short-Term Memory (LSTM) network, an improved bidirectional LSTM with additive attention, and a Transformer encoder with sinusoidal positional encoding. Each model receives a 96-hour multivariate history of seven load and temperature variables and predicts the next 24 hours of oil temperature (OT) in a multi-output, single-shot manner. Models are trained under a chronological 70/10/20 split, with scaler statistics fitted on training data only, and are evaluated by mean squared error (MSE), mean absolute error (MAE), and root mean squared error (RMSE) on the original temperature scale. The baseline LSTM achieved the best test performance (MSE 3.452, MAE 1.425, RMSE 1.858), while the attention-augmented LSTM (RMSE 1.938) and the Transformer (RMSE 1.934) were approximately 4% worse in RMSE. These results indicate that, for a modest-sized, strongly periodic single-source dataset, added architectural complexity does not guarantee better accuracy. We discuss plausible causes, including validation-loss instability, systematic over-prediction by the Transformer, and the limits of a single-run evaluation.**

***Index Terms*—time series forecasting, LSTM, attention mechanism, Transformer, ETTh1, oil temperature, predictive maintenance**

I. Introduction

Power transformers are among the most critical and costly assets in electrical grids, and their top-oil temperature is a key indicator of thermal stress, loading conditions, and insulation aging. Reliable short-term forecasts of oil temperature allow operators to schedule load, anticipate overheating, and plan maintenance before faults occur. Because oil temperature depends on load, ambient conditions, and thermal inertia, it exhibits daily and seasonal periodicity as well as irregular fluctuations, which makes it a demanding and practically relevant forecasting target [1].

Recurrent neural networks, in particular the Long Short-Term Memory (LSTM) network [2], have long been the default approach to sequence forecasting. More recently, attention mechanisms [3] and the Transformer architecture [4] have been adopted for long-horizon forecasting, motivated by their ability to model long-range dependencies in parallel. However, evidence on whether these more elaborate models consistently outperform well-tuned recurrent baselines is mixed, and several studies have questioned the benefit of Transformers on time series tasks [5].

This paper investigates this question on the ETTh1 benchmark [1]. We implement and compare three architectures under identical data preparation, optimizer, and loss: (i) a baseline two-layer LSTM, (ii) an improved bidirectional LSTM with an additive attention layer, and (iii) a Transformer encoder. The contributions of this work are as follows:

1) a reproducible multivariate-to-univariate 96-to-24-hour forecasting pipeline with leakage-free normalization;

2) a like-for-like comparison of recurrent, attention-augmented recurrent, and self-attention models using MSE, MAE, and RMSE on the original temperature scale; and

3) an analysis of why the simplest model prevailed, including training dynamics and prediction bias.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the dataset and preprocessing. Section IV details the model architectures. Section V presents the experimental setup, Section VI discusses the results, and Section VII concludes.

II. Related Work

LSTM networks mitigate the vanishing-gradient problem of simple recurrent networks through gated memory cells [2] and remain a strong baseline for forecasting. Bidirectional recurrent networks [6] process a sequence in both temporal directions, which is valid for the encoder over a fixed historical window. Additive attention [3] lets a model weight hidden states across the input window rather than relying solely on the final state.

The Transformer [4] replaces recurrence with multi-head self-attention. Specialized variants such as Informer [1], Autoformer [7], and PatchTST [8] address the quadratic cost of attention and the loss of temporal ordering information in long-horizon forecasting. Zeng et al. showed that simple linear models can rival or outperform many Transformer-based forecasters on standard benchmarks [5], underscoring the importance of strong baselines. Our study follows this spirit by comparing a vanilla Transformer encoder against recurrent models under identical conditions.

III. Dataset and Preprocessing

A. ETTh1 Dataset

We use the ETTh1 subset of the Electricity Transformer Temperature (ETT) dataset [1], which contains hourly records from an electric transformer over approximately two years. The dataset has 17,420 samples and eight columns: a

timestamp and seven numeric variables, namely six power-load features—High/Middle/Low UseFul Load (HUFL, MUFL, LUFL) and High/Middle/Low UseLess Load (HULL, MULL, LULL)—and the target, oil temperature (OT). No missing values were present. The target has a mean of 13.32, a standard deviation of 8.57, and a range of −4.08 to 46.01 (Table I).

TABLE I
SUMMARY OF THE ETTH1 DATASET

Property	Value
Records (hourly)	17,420
Input variables	7 (6 load + OT)
Target variable	OT (oil temperature)
Missing values	0
OT mean / std	13.32 / 8.57
OT min / max	−4.08 / 46.01

B. Splitting, Normalization, and Windowing

Data were sorted chronologically and divided without shuffling into training (70%, 12,194 samples), validation (10%, 1,742), and test (20%, 3,484) sets. A standard scaler was fitted exclusively on the training partition and applied to all partitions to prevent information leakage. Sliding windows were then generated with an input length of 96 hours (all seven variables) and a prediction length of 24 hours (OT only). This yields 12,075 training, 1,719 validation, and 3,461 test windows, each with input shape 96×7 and target shape 24. For all reported metrics, predictions and ground truth were inverse-transformed to the original temperature scale.

IV. MODEL ARCHITECTURES

All models map the input window $X \in \mathbb{R}^{(96\times7)}$ to a 24-dimensional forecast vector in a single forward pass, avoiding autoregressive error accumulation.

A. Baseline LSTM

The baseline stacks two LSTM layers with 128 hidden units and inter-layer dropout of 0.2. The hidden state at the final time step is passed to a fully connected layer producing the 24 outputs. The model has approximately 0.21 million trainable parameters.

B. Improved LSTM with Attention

The improved model replaces the unidirectional encoder with a two-layer bidirectional LSTM (128 units per direction, dropout 0.3), producing 256-dimensional states at every time step. An additive attention module (256→64→1 with tanh activation) scores each step; scores are normalized by a softmax over time and used to form a weighted context vector, which is regularized by dropout and mapped to the 24 outputs by a linear layer. This variant has approximately 0.56 million parameters and uses a weight decay of 10^{-5}.

C. Transformer Encoder

The Transformer linearly projects the seven input features into a 128-dimensional embedding, adds fixed sinusoidal positional encodings [4], and applies two encoder layers with four attention heads, a 256-dimensional feed-forward sublayer, GELU activation, and dropout of 0.2. The representation of the last time step is passed through dropout and a linear head to produce the forecast. The model has approximately 0.27 million parameters.

V. EXPERIMENTAL SETUP

A. Training Configuration

All models were implemented in PyTorch [9] and trained on a single NVIDIA Tesla T4 GPU with the Adam optimizer [10] (learning rate 10^{-3}) and an MSE loss on normalized targets. Mini-batches of 64 windows were used, with gradient-norm clipping at 1.0 [11] in the primary runs. Early stopping on validation loss with a patience of 3–5 epochs (maximum 15–30 epochs depending on the model) was used, and the weights with the lowest validation loss were restored for testing. Table II lists the main settings.

TABLE II
MODEL AND TRAINING SETTINGS

Setting	Baseline	Improved	Transf.
Layers	2	2 (BiLSTM)	2
Hidden / d_model	128	128×2	128
Dropout	0.2	0.3	0.2
Weight decay	0	10^{-5}	10^{-5}
Attention	—	Additive	4-head SA
Params (M)	0.21	0.56	0.27

B. Evaluation Metrics

Performance is measured on the test set, over all 24 horizon steps, with N the total number of predicted values:

$$MSE = (1/N)\,\Sigma\,(y_i - \hat{y}_i)^2 \quad (1)$$

$$MAE = (1/N)\,\Sigma\,|y_i - \hat{y}_i| \quad (2)$$

$$RMSE = \sqrt{MSE} \quad (3)$$

Lower values indicate better accuracy. RMSE is expressed in the same unit as OT and penalizes large errors more heavily than MAE.

VI. RESULTS AND DISCUSSION

A. Quantitative Comparison

Table III and Fig. 1 report the test-set errors. The baseline LSTM achieved the lowest error on all three metrics (MSE 3.452, MAE 1.425, RMSE 1.858). Relative to the baseline, the improved LSTM with attention had 8.8% higher MSE, 5.1% higher MAE, and 4.3% higher RMSE, while the Transformer had 8.4% higher MSE, 6.6% higher MAE, and 4.1% higher RMSE. The two more complex models performed almost identically in MSE and RMSE, with the Transformer showing the larger MAE.

TABLE III
TEST-SET FORECASTING PERFORMANCE

Model	MSE	MAE	RMSE
Baseline LSTM	3.4520	1.4252	1.8580
Improved LSTM + Attn.	3.7545	1.4979	1.9376
Transformer	3.7411	1.5190	1.9342

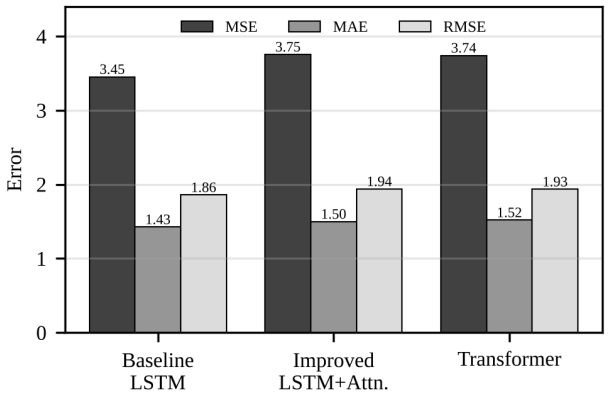


Fig. 1. Test-set MSE, MAE, and RMSE for the three forecasting models.

B. Qualitative Forecast Behavior

Fig. 2 plots the first 30 predicted values against ground truth. On this segment, the baseline LSTM tracks the level of OT most closely (mean signed error +0.23, MAE 0.72), followed by the improved LSTM (+0.55, 0.85). The Transformer exhibits a persistent positive offset, over-predicting every one of the 30 samples by 1.54 on average. Notably, all models respond to the sharp temperature excursion around samples 20–24 with a delayed or overshooting response, a typical failure mode of single-shot forecasters faced with abrupt regime changes. These per-segment figures are illustrative; the full-test-set metrics in Table III are the authoritative comparison.

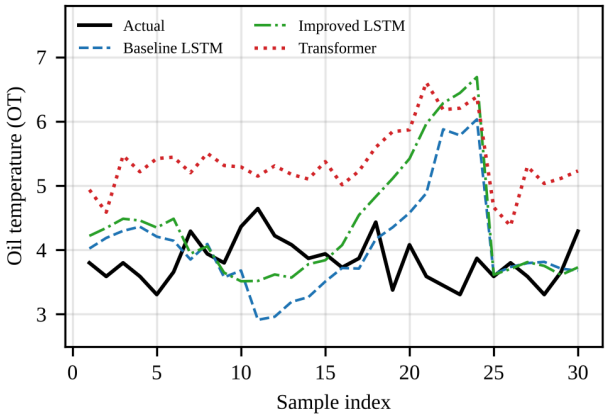


Fig. 2. Actual versus predicted oil temperature for the first 30 test samples.

C. Training Dynamics

Training loss decreased steadily for all models, whereas validation loss was volatile and not monotonic, frequently rising while training loss continued to fall. Because the validation partition covers a single contiguous 10% period (about 1,742 hours), seasonal and load-regime differences between training and validation periods can produce such oscillations, and early stopping may therefore select an epoch partly by chance. This noise plausibly affects the more flexible models more strongly, as they fit the training distribution faster.

D. Discussion

Three factors may explain why the added complexity did not help. First, ETTh1 is comparatively small and dominated by daily periodicity, which a two-layer LSTM can capture adequately; additional capacity brings more overfitting risk than benefit. Second, both complex models pool information differently (attention-weighted context and last-token readout), and the Transformer's readout from a single token may discard useful temporal context. Third, the vanilla Transformer lacks the inductive biases (decomposition, patching, locality) that specialized forecasting architectures employ [7], [8]. These explanations are hypotheses consistent with the observations rather than conclusions established by ablation.

E. Limitations

The findings are subject to several limitations. Results come from a single training run per model without fixed random seeds, so differences of 4–9% may fall within run-to-run variance. Only one dataset subset, one horizon (24 h), and one hyperparameter configuration per model were evaluated, with no systematic tuning. Statistical significance testing and comparison with non-neural baselines (e.g., persistence, ARIMA, or linear models) were not performed. Future work should include multiple seeds with confidence intervals, rolling-origin validation, additional ETT subsets, and stronger baselines.

VII. Conclusion

This paper compared a baseline LSTM, a bidirectional LSTM with attention, and a Transformer encoder for 24-hour oil-temperature forecasting on ETTh1. Under identical preprocessing and training protocols, the simple baseline LSTM achieved the best accuracy (RMSE 1.858), outperforming the attention-augmented LSTM (1.938) and the Transformer (1.934). The results reinforce the value of strong, simple baselines for small, periodic industrial time series and caution against assuming that architectural sophistication yields better forecasts. Future work will pursue multi-seed evaluation, longer horizons, specialized forecasting Transformers, and decomposition-based inputs.

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